JOSEPH MCKENNA

Backend Engineer | Python | AWS Serverless & Event-Driven Systems
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PROFESSIONAL SUMMARY

Backend engineer specializing in Python and AWS-based distributed systems. Experienced designing and operating event-driven architectures that process thousands of healthcare transactions daily using Lambda, SQS, and Step Functions. Skilled in building cloud data pipelines, document processing platforms, and fault-tolerant automation systems that transform unstructured data into production workflows. Currently pursuing a Master's in Artificial Intelligence & Machine Learning with a focus on scalable analytics and intelligent automation systems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
Frameworks: Django, Django REST Framework, React
Databases: PostgreSQL, DynamoDB, SQL Server
Cloud & Infrastructure: AWS (Lambda, SQS, SNS, Step Functions, S3, CloudWatch, Textract), Terraform, Docker
Architecture: Serverless architecture, event-driven systems, distributed processing, idempotent workflows, asynchronous pipelines
Data & Standards: ANSI X12 (837D, 837I, 837P), EDI transaction processing, OCR data extraction pipelines

PROFESSIONAL EXPERIENCE

Gainwell Technologies — Remote **May 2022 - Present**
Backend Engineer | Professional Application Designer
Medicare Claims Processing Platform

• Designed and built a distributed healthcare document processing platform that automates the ingestion, extraction, validation, and transformation of healthcare claims into compliant X12 EDI transactions.
• Architected event-driven backend systems using Python and AWS (Lambda, SQS, Step Functions) supporting Medicaid claims processing for the West Virginia state government across all claim types processing 2,000+ Medicaid claims daily in a fault-tolerant distributed pipeline with automated retries, dead-letter queues, and idempotent workflows.
• Built an OCR document processing pipeline using AWS Textract to convert unstructured TIFF claim documents into structured claim data with automated confidence scoring.
• Developed an internal Python Tkinter validation interface enabling human review of low-confidence OCR fields, improving data accuracy before claim generation.
• Implemented automated FTP ingestion and routing workflows to organize and batch incoming documents by claim type and processing date.
• Designed mapping and validation engines to transform extracted claim data into ANSI X12 837 transactions for production submission.
• Eliminated a 30,000-document processing backlog by converting manual workflows into automated distributed processing systems.
• Reduced reliance on offshore contractors and generated $250K in annual cost savings through automation and internal platform development.
• Implemented centralized logging, monitoring, and alerting using AWS CloudWatch to improve system reliability and reduce incident resolution time.

PERSONAL PROJECTS

22 to LIFE — Python, Django, React, AWS (In Development)
Full-stack platform supporting secure community interaction for veterans and first responders.

• Architecting backend services using Django REST Framework to manage authentication, messaging, and user profile workflows.
• Developing a React frontend supporting real-time user interaction and protected application routes.
• Designing relational database schemas to support scalable user-generated content and community engagement features.

• Implementing role-based access controls and data validation to ensure secure handling of sensitive user information.

---------- **EDUCATION**----------

Master of Science — Artificial Intelligence & Machine Learning (In Progress) — Colorado State University Global
Bachelor of Science — Criminal Justice — Briarcliffe College
Full Stack Software Development Boot Camp — The Tech Academy